

D316

IT Foundations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A help desk technician at a dental office with nine chairs gets a ticket that the front-desk PC will not print. The office manager says the PC is only a year old and cost \$1,200. What should the technician do first?

- A. Reinstall the printer driver.
- B. Identify the problem by questioning the user and noting recent changes.
- C. Document the fix in the ticket system.
- D. Replace the PC, since it cost \$1,200.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A technician has implemented a fix and the user can print again. What should the technician do next?

- A. Escalate to a senior technician.
- B. Close the ticket without further checks.
- C. Verify full system functionality and, if applicable, implement preventive measures.
- D. Establish a theory of probable cause.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A small architecture firm buys four 2 TB drives for its file server from a vendor that offers free shipping. It wants the array to survive one drive failure and to give the most usable space possible. Which RAID level and usable capacity fit?

- A. RAID 0 with 8 TB usable
- B. RAID 10 with 4 TB usable
- C. RAID 5 with 6 TB usable
- D. RAID 5 with 8 TB usable

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A RAID 5 array provides 12 TB of usable space and is built from 3 TB drives. How many drives are in the array?

- A.** 3
- B.** 5
- C.** 8
- D.** 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A remote worker at a marketing agency with 45 employees can browse the web but cannot use Remote Desktop to reach her office PC. Her home router is a new Wi-Fi 6 model. Which port should the firewall administrator check?

- A.** 443
- B.** 22
- C.** 3389
- D.** 23

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A firewall log shows allowed traffic on TCP port 143. Which service is most likely being used?

- A.** IMAP email retrieval
- B.** SMTP email sending
- C.** POP3 email retrieval
- D.** DNS

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A user at a credit union branch, whose laptop was reimaged last week, cannot reach any internal or internet sites. ipconfig shows 169.254.33.10. What is the most likely cause?

- A.** The reimage installed the wrong browser.
- B.** The laptop could not get an address from a DHCP server.
- C.** The DNS server is down.
- D.** The address is valid; 169.254 is a normal private range.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A technician wants a workstation to show a 169.254.x.x address as a test. Which condition would cause it?

- A.** The default gateway is wrong.
- B.** DNS returns the wrong address for a website.
- C.** The NIC has a static IP of 192.168.1.50.
- D.** The NIC is set to obtain an address automatically, and the DHCP server is unreachable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A law firm's laser printer, bought last year for \$2,400, produces pages where the toner smudges and wipes off when touched. The toner cartridge was replaced two weeks ago. Which part is most likely failing?

- A.** The imaging drum
- B.** The toner cartridge
- C.** The pickup rollers
- D.** The fuser assembly

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A technician at a school with 700 students is replacing a failed power supply in a desktop. The school's IT budget was cut by 10% this year. The old PSU is still under warranty. What is the correct safety practice?

- A.** Keep the PC plugged in so it stays grounded while working.
- B.** Open the PSU and replace only the failed capacitor to save money.
- C.** Once unplugged, a PSU is always safe to open.
- D.** Unplug the PC, use ESD protection, and replace the PSU as a whole unit without opening it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A video editor at a small studio installs a new M.2 SSD bought on sale for \$90, but benchmark speeds match an older SATA SSD. The motherboard manual says the M.2 slot supports SATA only. The editor works on 4K footage. Why is it slow?

- A.** The slot runs at SATA speed, so even a fast drive is limited to the SATA interface.
- B.** SSDs are slower than HDDs for large files.
- C.** 4K footage is too large for any SSD.
- D.** All M.2 drives are NVMe, so the drive must be faulty.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A family-run hotel with 22 rooms has a single dual-band access point in the lobby. Guests in rooms at the far end of the building, behind several walls, keep losing the 5 GHz network. The owner recently repainted the halls. Which change most likely improves their connection?

- A.** Connect those guests to the 2.4 GHz band.
- B.** Repaint with a lighter color to help the signal.
- C.** Force all devices onto 5 GHz for more speed.
- D.** Switch to 5 GHz because it always has better coverage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A help desk technician at a dental office ~~with nine chairs~~ gets a ticket that the front-desk PC will not print. ~~The office manager says the PC is only a year old and cost \$1,200.~~ What should the technician do first?

Target word: first troubleshooting step **Decides:** The method starts by identifying the problem.

A	Dropped element	Jumps to a fix, skipping identification and theory.
B	Correct	Step one is gathering information and identifying symptoms.
C	Inverted operation	Documentation is the last step.
D	Camouflage	Price and age do not decide the first step.

2. Answer C Concept 1 • transfer

A technician has implemented a fix and the user can print again. What should the technician do next?

Target word: step after implementing the solution **Decides:** Verify full functionality and add preventive measures before documenting.

A	Adjacent category	Escalation fits when a theory cannot be confirmed, not after a working fix.
B	Dropped element	Skips verification and documentation.
C	Correct	Verification comes right after implementation.
D	Inverted operation	That is step two, already done.

3. Answer C Concept 2

A small architecture firm buys four 2 TB drives for its file server ~~from a vendor that offers free shipping.~~ It wants the array to survive one drive failure and to give the most usable space possible. Which RAID level and usable capacity fit?

Target word: RAID level and capacity **Decides:** Four drives, one-drive fault tolerance, maximize space.

A	Confusion set	Maximum space but no fault tolerance.
B	Adjacent category	Survives a failure but gives less space than RAID 5.
C	Correct	Parity costs one drive: $(4 - 1) \times 2 \text{ TB} = 6 \text{ TB}$.
D	Dropped element	Leaves out the drive's worth of capacity used for parity.

4. Answer B Concept 2 • transfer

A RAID 5 array provides 12 TB of usable space and is built from 3 TB drives. How many drives are in the array?

Target word: solve for drive count **Decides:** Usable = $(n - 1) \times \text{drive size}$.

A	Wrong stem number	The drive size, not the count.
B	Correct	$12 / 3 = 4$ data drives, plus 1 for parity = 5.
C	Confusion set	Applies the mirroring rule (half capacity) instead of parity.
D	Dropped element	Forgets the parity drive.

5. Answer C Concept 3

A remote worker at a marketing agency ~~with 45 employees~~ can browse the web but cannot use Remote Desktop to reach her office PC. ~~Her home router is a new Wi-Fi 6 model.~~ Which port should the firewall administrator check?

Target word: RDP port **Decides:** Remote Desktop uses TCP 3389.

A	Adjacent category	HTTPS; web browsing already works.
B	Confusion set	SSH, another remote access protocol.
C	Correct	RDP listens on 3389.
D	Inverted operation	Telnet, an insecure text remote shell, not RDP.

6. Answer A Concept 3 · transfer

A firewall log shows allowed traffic on TCP port 143. Which service is most likely being used?

Target word: reverse: port to protocol **Decides:** 143 is IMAP.

A	Correct	IMAP uses port 143.
B	Adjacent category	SMTP uses 25.
C	Confusion set	POP3 uses 110.
D	Inverted operation	DNS uses 53.

7. Answer B Concept 4

A user at a credit union branch, ~~whose laptop was reimaged last week,~~ cannot reach any internal or internet sites. ipconfig shows 169.254.33.10. What is the most likely cause?

Target word: meaning of a 169.254 address **Decides:** APIPA means DHCP could not be reached.

A	Camouflage	The reimage is a distraction; the address points to DHCP.
B	Correct	Windows assigns 169.254.x.x when DHCP fails.
C	Adjacent category	DNS failure still leaves a valid DHCP address.
D	Confusion set	Private ranges are 10.x, 172.16-31.x, and 192.168.x; 169.254 is link-local.

8. Answer D Concept 4 · transfer

A technician wants a workstation to show a 169.254.x.x address as a test. Which condition would cause it?

Target word: reverse: condition that produces APIPA **Decides:** DHCP is set but no server answers.

A	Adjacent category	A bad gateway blocks remote traffic but does not change the address.
B	Confusion set	Name resolution does not assign host addresses.
C	Inverted operation	Static addressing never falls back to APIPA.
D	Correct	That is exactly when APIPA self-assigns.

9. Answer D Concept 5

A law firm's laser printer, ~~bought last year for \$2,400~~, produces pages where the toner smudges and wipes off when touched. ~~The toner cartridge was replaced two weeks ago~~. Which part is most likely failing?

Target word: part responsible for bonding toner **Decides:** Toner not fused to the paper.

A	Adjacent category	Drum faults cause ghosting or lines, not smearing.
B	Camouflage	A recent cartridge swap does not explain unfused toner.
C	Confusion set	Pickup rollers cause paper feed problems.
D	Correct	The fuser melts and presses toner onto the paper.

10. Answer D Concept 6

A technician at a school ~~with 700 students~~ is replacing a failed power supply in a desktop. ~~The school's IT budget was cut by 10% this year. The old PSU is still under warranty~~. What is the correct safety practice?

Target word: safe handling of a power supply **Decides:** PSUs hold charge in capacitors; replace the whole unit.

A	Inverted operation	Grounding comes from an ESD strap and mat, not live power.
B	Camouflage	Budget pressure does not make opening a PSU safe.
C	Absolute language	Capacitors hold charge after power is removed.
D	Correct	Capacitors can hold a dangerous charge; PSUs are field-replaceable units.

11. Answer A Concept 7

A video editor at a small studio installs a new M.2 SSD ~~bought on sale for \$90~~, but benchmark speeds match an older SATA SSD. The motherboard manual says the M.2 slot supports SATA only. ~~The editor works on 4K footage~~. Why is it slow?

Target word: cause of SATA-level speed **Decides:** The slot supports only the SATA interface.

A	Correct	M.2 is a form factor; the interface set by the slot decides speed.
B	Inverted operation	SSDs are faster than HDDs.
C	Camouflage	File size does not explain an interface limit.
D	Absolute language	M.2 drives and slots may be SATA or NVMe.

12. Answer A Concept 8

A family-run hotel ~~with 22 rooms~~ has a single dual-band access point in the lobby. Guests in rooms at the far end of the building, behind several walls, keep losing the 5 GHz network. ~~The owner recently repainted the halls~~. Which change most likely improves their connection?

Target word: frequency choice for range **Decides:** Distance and walls favor 2.4 GHz.

A	Correct	2.4 GHz travels farther and passes through walls better.
B	Camouflage	Paint color is irrelevant.
C	Inverted operation	5 GHz has shorter range, the source of the problem.
D	Absolute language	5 GHz trades range for speed.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Troubleshooting methodology order	Identify the problem, establish a theory, test the theory, establish a plan and implement it, verify full functionality with preventive measures, then document.	Jumping to a fix before identifying the problem, or skipping verification and documentation.	"What should the technician do first / next?"
2	RAID levels and capacity	RAID 0 stripes with no redundancy; RAID 1 mirrors (half capacity); RAID 5 uses parity with at least 3 drives and loses one drive of capacity; RAID 10 mirrors stripes with at least 4 drives and half capacity.	Calling RAID 0 fault tolerant, or counting all drives as usable in RAID 5.	A drive count, drive size, and a need for speed or fault tolerance.
3	Common ports and protocols	FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, HTTP 80, POP3 110, IMAP 143, HTTPS 443, SMB 445, RDP 3389.	Swapping POP3 and IMAP, or SSH and Telnet.	A service that works or fails and a firewall rule.
4	APIPA addresses	A Windows host with a 169.254.x.x address could not reach a DHCP server and assigned itself a link-local address.	Blaming DNS when the address is 169.254.x.x.	An address starting with 169.254.
5	Laser printer symptoms	Toner that smears or rubs off means the fuser is not bonding it; repeating ghost images point to the drum or cleaning; vertical lines often mean a scratched drum.	Replacing the toner cartridge for a fusing problem.	Toner wipes off the page.
6	Safety and ESD	Power down and unplug before opening a PC, use an ESD strap and mat, never open a power supply or CRT because capacitors hold charge, and check the SDS for chemical handling.	Thinking an unplugged power supply is safe to open.	A technician about to open or clean a component.
7	Storage types and interfaces	SSDs have no moving parts and are faster than HDDs; NVMe SSDs use PCIe lanes and are faster than SATA SSDs; an M.2 slot may support SATA, NVMe, or both.	Assuming every M.2 drive is NVMe.	An upgrade that is slower than expected or does not fit the slot.
8	Wireless frequency trade-offs	2.4 GHz reaches farther and penetrates walls better but is slower and more crowded; 5 GHz is faster with more channels but shorter range.	Thinking 5 GHz gives better range.	Coverage at a distance versus speed near the access point.

Course scope (public description): IT Foundations provides learners with an understanding of personal computer components and their functions in a desktop system; a knowledge of computer data storage and retrieval; and skills in classifying, installing, configuring, optimizing, upgrading, and troubleshooting printers, laptops, portable devices, operating systems, networks, and system security. This course also gives learners the ability to recommend appropriate tools, diagnostic procedures, preventative maintenance, and troubleshooting techniques for personal computer components in a desktop system; strategies for identifying, preventing, and reporting safety hazards and environmental or human accidents in technological environments; and effective communication skills for interacting with colleagues and clients, including job-related professional behavior. The course prepares learners for the CompTIA A+ Core 1 certification exam.